

Simple Manual PWM Controller

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The pulse width modulation (PWM) is used to control electrical loads such as incandescent lamps, LEDs and DC motors. Presented here is a simple manual PWM controller that makes use of IC 74C14 having six inverters with Schmitt triggers to generate PWM signals. You may also use 74HC14, 74AC14 and similar other ICs in place of 74C14 to produce PWM signals.

Circuit and working

Fig. 1 shows the circuit of a simple manual PWM controller built around hex inverter IC 74C14 (IC1), and Dar-

lington pair transistor TIP122 (T1). The figure also shows different loads that can be connected to CON2 for control through PWM signals.

An oscillator is made using gate N1, the frequency of which can be selected through rotary switch S1. It is preferably to limit the working frequency to well below 30kHz. The limitation comes from the switching speed on transistor T1. Duration of the output pulses is adjusted with potentiometer VR1. Gates N2 through N6 are used to increase the output current of IC1 needed for driving transistor T1. IC 74C14 can provide a few milliamperes of output current and its power supply range is from 3V to 15V. 74C14

can provide $\pm 24\text{mA}$ per inverter but its power supply range is only from 2V to 6V.

The output current of IC1 is amplified by the transistor, which can be a simple transistor such as BD135, TIP41, 2SD882 or 2N3055. It should preferably be a Darlington transistor such as TIP122, BD681 or BDW93. The output current for the load connected across CON2 depends on transistor T1. Fuse F1 protects the transistor.

Diodes D1 through D4 provide over-current protection for transistor

Test Points

Test point	Details
TP0	GND
TP1	12V
TP2	Pulses with frequency selected by switch S1

PARTS LIST

Semiconductors:

- IC1 - 74C14 hex inverter
 - D1-D6 - 1N4001 rectifier diode
 - D7-D8 - 1N4148 signal diode
 - T1 - TIP122 power transistor
 - LED1 - 5mm green LED
 - LED2 - 5mm red LED
 - LED3-LED26 - 5mm white bright LED
- Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):
- R1, R2 - 4.7-kilo-ohm
 - R3, R5, R8 - 1-kilo-ohm
 - R4 - 10-kilo-ohm
 - R6 - 0.22-ohm, 2W
 - R7 - 1-ohm, 2W
 - R9-R14 - 220-ohm
 - VR1 - 100-kilo-ohm potentiometer

Capacitors:

- C1 - 1 μF , 16V electrolytic
- C2 - 100nF ceramic disk
- C3 - 10nF ceramic disk
- C4 - 1nF ceramic disk
- C5 - 0.33 μF , 25V electrolytic
- C6 - 1000 μF , 25V electrolytic

Miscellaneous:

- F1 - 2A fuse
- S1 - 4-way rotary switch
- CON1-CON6 - 2-pin connector
- S2, S3 - On/off switch
- B1, B2 - 12V, 5W bulb
- M1 - 12V DC motor
- 14-pin IC base
- 12V, 2A regulated supply

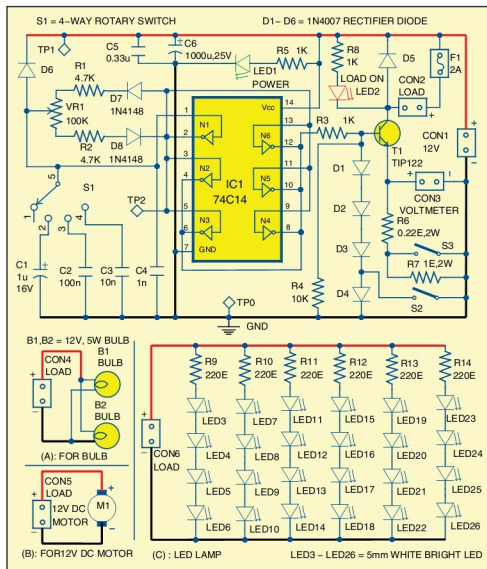


Fig. 1: Circuit diagram for simple manual PWM controller